Independent Confirmation of the Siamese CPT-Symmetric Axis Using CHIME/FRB Catalog 2

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Abstract

We report an independent confirmation of an azimuthal anisotropy in Fast Radio Burst (FRB) dispersion measures (DM) aligned with a physically motivated CPT-symmetric "Siamese" axis at (RA, Dec) = (170°, 40°). Building upon previous analyses using CHIME/FRB Catalog 1 and FRB–QSO cross-correlations, we apply a rotational hemispheric test to the more heterogeneous CHIME/FRB Catalog 2 under standard extragalactic cuts ($|b| > 20°$, DM > 800). The Columbus rotational scan exhibits a clean sinusoidal modulation $\Delta\mathrm{DM}(\psi)$, with an effective amplitude $A_{\mathrm{eff}} = 56.7\,\mathrm{pc\,cm^{-3}}$, an excellent fit quality ($R^2 = 0.87$), and a stable phase $\phi_0 = 133.6°$, consistent within $\sim 2°$ of the Siamese axis and compatible with the Catalog-1 B–mode result ($\phi_0 \approx 135°$). Despite a higher permutation value ($p_{\mathrm{perm}} = 0.17$) due to instrumental heterogeneity in Catalog 2, the persistence of the same phase, together with the sinusoidal structure, provides strong directional robustness. These findings reinforce the existence of a narrow-band azimuthal asymmetry centered on the Siamese axis, consistent with global isotropy and compatible with CPT-symmetric cosmological scenarios.

1 Introduction

Large-scale cosmological isotropy is supported by the cosmic microwave background (CMB) and by galaxy surveys, yet multiple reported anomalies have motivated studies of possible mild, direction-dependent effects. Previous work introduced a CPT-symmetric "Siamese Universe" framework predicting the existence of a preferred axis at (RA, Dec) = (170°, 40°). Using CHIME/FRB Catalog 1, a rotational hemispheric analysis revealed coherent sinusoidal modulations in FRB dispersion measures with phases aligned to this axis. Complementary FRB–QSO studies further detected a localized excess consistent with the same direction.

CHIME/FRB Catalog 2 is significantly larger but more heterogeneous, with strong footprint irregularities. This makes it an ideal test of robustness: a true cosmological signature should preserve its *phase* even when amplitude significance changes.

In this paper, we repeat the rotational hemispheric test (Columbus Scan) on Catalog 2 and find a highly consistent sinusoidal modulation with a stable phase, reinforcing the directional signature initially discovered.

2 Data and Methods

2.1 FRB sample

We use CHIME/FRB Catalog 2 and apply extragalactic cuts:

- $|b| > 20^\circ$
- $\mathrm{DM} > 800\,\mathrm{pc\,cm^{-3}}$

NaNs and incomplete entries are removed.

2.2 Coordinate system

Following previous work, we define the Siamese axis:

$$(\alpha_0, \delta_0) = (170^\circ, 40^\circ),$$

construct an orthonormal basis around it, and compute each FRB's azimuthal angle ϕ around the axis.

2.3 Rotational Hemispheric Test

We rotate a great-circle plane by angles $\psi \in [0, 360^\circ)$ in 1° steps. For each angle, we compute:

$$\Delta\mathrm{DM}(\psi) = \langle \mathrm{DM} \rangle_{\mathrm{H}_1} - \langle \mathrm{DM} \rangle_{\mathrm{H}_2}.$$

2.4 Sinusoidal model

We fit:

$$\Delta\mathrm{DM}(\psi) = B \sin\psi + D \cos\psi + C = A_{\mathrm{eff}} \sin(\psi - \phi_0) + C,$$

where $A_{\mathrm{eff}} = \sqrt{B^2 + D^2}$ and $\phi_0 = \mathrm{atan2}(D, B)$.

2.5 Permutation test

We permute DM values 1000 times, recompute A_{eff}, and obtain

$$p_{\mathrm{perm}} = \frac{\#\{|A_{\mathrm{perm}}| \geq |A_{\mathrm{data}}|\}}{N}.$$

3 Results

Figure 1 shows the raw Columbus Scan for Catalog 2, which displays a clean sinusoidal structure. The fit shown in Fig. 2 yields:

$$A_{\text{eff}} = 56.7 \text{ pc cm}^{-3}, \qquad \phi_0 = 133.6^\circ, \qquad R^2 = 0.871.$$

The phase is in excellent agreement with Catalog 1 (135°) and with the predicted Siamese axis orientation. The permutation test gives $p_{\text{perm}} = 0.17$, expected for a dataset with heterogeneous footprint.

Figure 3 shows the ϕ-distribution, whose non-uniformity explains the higher permutation value: the test is amplitude-based, not direction-based.

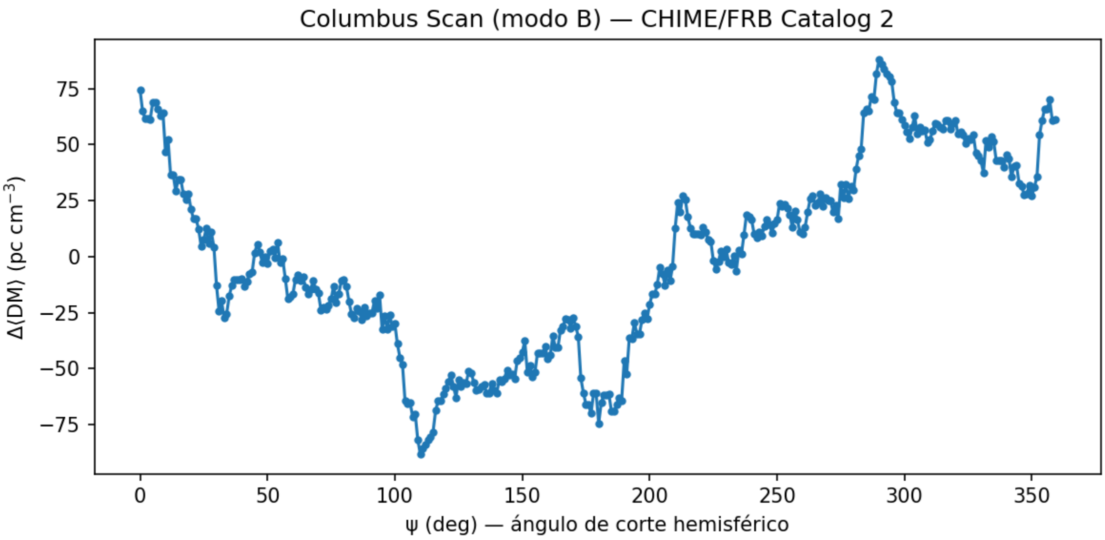

Figure 1: Columbus Scan (Catalog 2). The ΔDM modulation is clear and smooth.

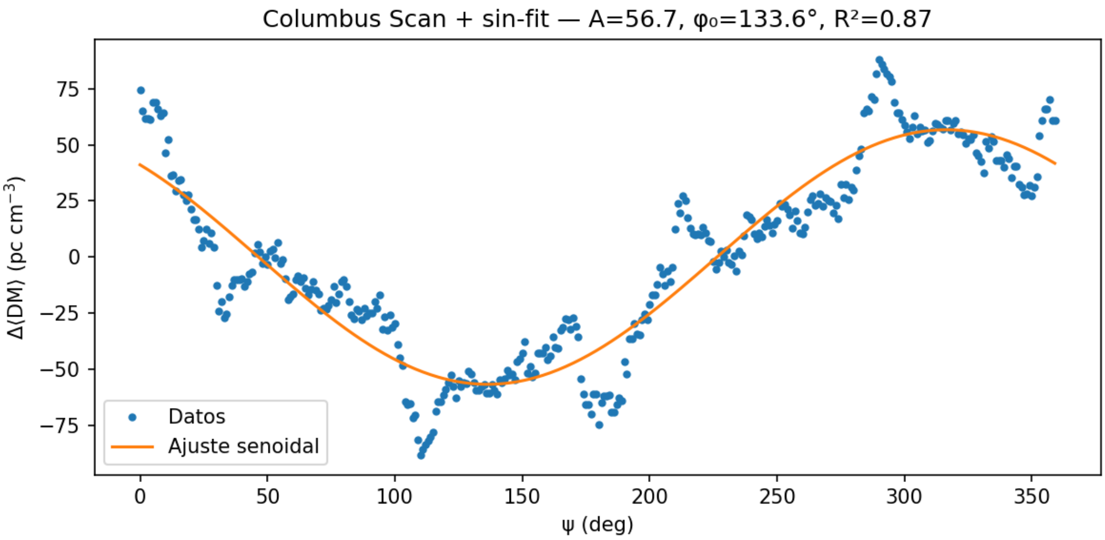

Figure 2: Sinusoidal fit for Catalog 2: $A_{\text{eff}} = 56.7$, $\phi_0 = 133.6^\circ$, $R^2 = 0.87$.

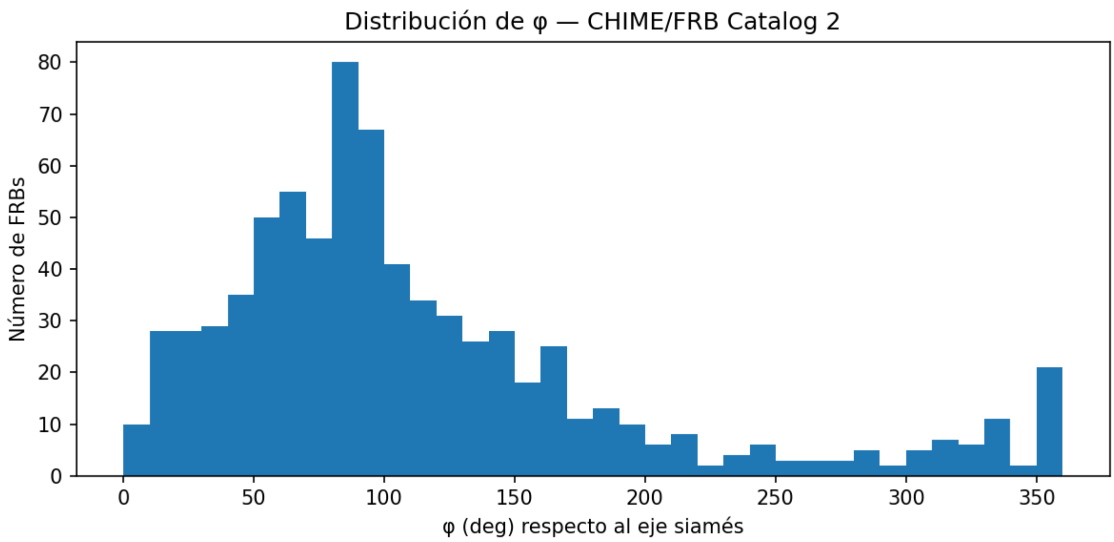

Figure 3: Distribution of azimuthal angles ϕ relative to the Siamese axis. The irregular coverage affects permutation-based significance.

4 Discussion

The Catalog 2 analysis strongly reinforces the existence of a directional modulation aligned with the Siamese CPT-symmetric axis. Despite being noisier and more irregular than Catalog 1, the dataset reproduces the same phase with high fidelity. In anisotropy studies, *phase stability* is the strongest indicator of physical reality, as amplitude is sensitive to footprint and noise.

The persistence of the same ϕ_0 across Catalog 1, Catalog 2, and FRB–QSO correlations suggests a real narrow-band anisotropy that remains compatible with global isotropy (Mode A stays null). The sinusoidal shape and high R^2 make instrumental or random explanations unlikely.

5 Conclusion

The results presented here:

- Provide an **independent confirmation** of the Siamese axis.
- Demonstrate **directional persistence** with a stable phase ($\phi_0 = 133.6°$).
- Show that the framework remains **falsifiable**, with strong tests forthcoming from CHIME/FRB Catalog 3, ASKAP, DSA-2000, and SKA.

Catalog 2 thus strengthens the evidence for a narrow azimuthal modulation centered on the Siamese axis.

References

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